

Immune System 1 & 2

1. Introduction
2. Organs & tissues
3. Nonspecific immunity / innate (*born with it*)
4. Overview of cells & chemicals (*mediators*)
5. Specific immunity / adaptive (*acquired*)
6. Immunity & health

Definitions

Immune: “freedom from”

Immune system: includes cells, processes & chemicals that keep us **disease** free

What causes **disease**?

1. **Pathogens:** bacteria, viruses & parasites
2. **Cancer cells**

Immunity: ability to recognize & eliminate foreign invaders or abnormal “*self*” cells that cause **disease**.

Immunology: study immune processes

Immune System: Big Picture FUNCTION

1. Keep out invaders

- External body surface - cutaneous membrane
- Internal body surface - mucous membranes

2. If they get in

- Eat them → cells phagocytize
- Shoot them → cells secrete chemicals or inject them with chemicals

2 Divisions of the Immune System

Nonspecific

- **Innate** – born with it
- **Involves:** WBCs (*not lymphocytes*), chemicals, processes that attack **ANY** bacteria, virus, virally infected cell, cancer cell

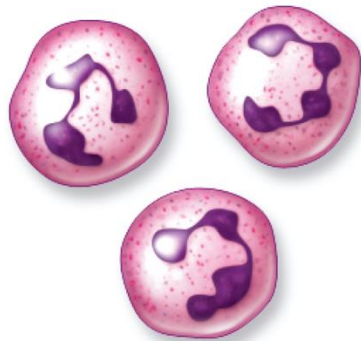
Specific

- **Acquired** – built through exposure to pathogens
- **Involves:** ONLY **B & T lymphocytes**
- Clones of **B & T lymphocytes**
 - Ability to recognize ALL foreign antigens; 1 clone/pathogen
 - 1 clone will multiply when stimulated
 - Produce “specific” immune response to “specific” pathogens

Formed Elements

*Cells of nonspecific immunity

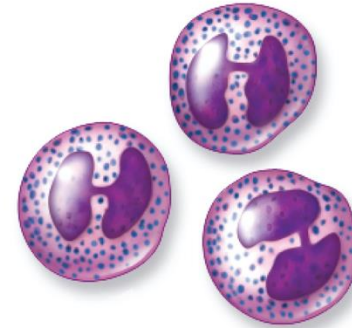
**Cells of specific immunity



*Neutrophils



*Eosinophils



*Basophils



**Lymphocytes



*
**Monocytes



Platelets



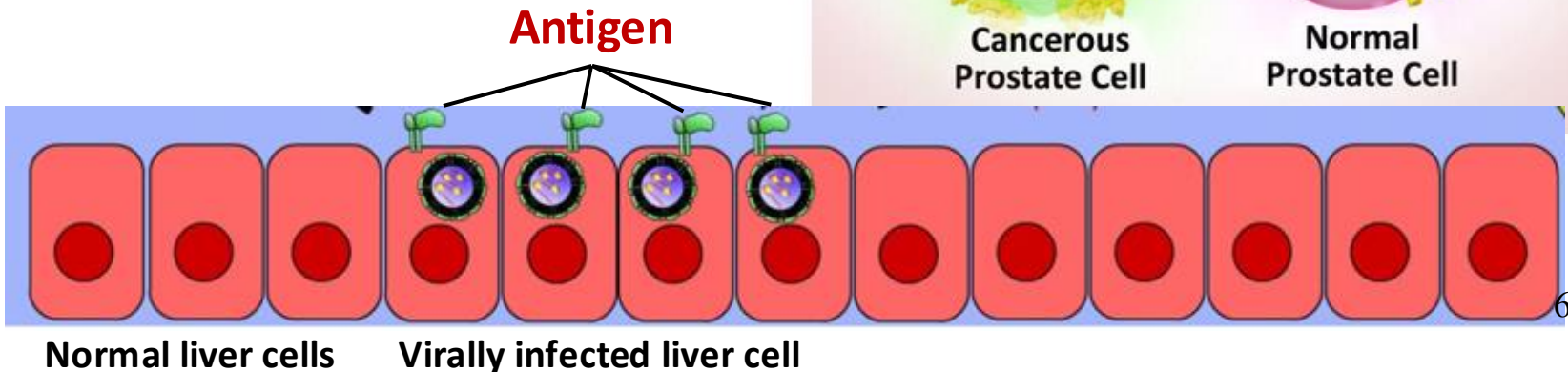
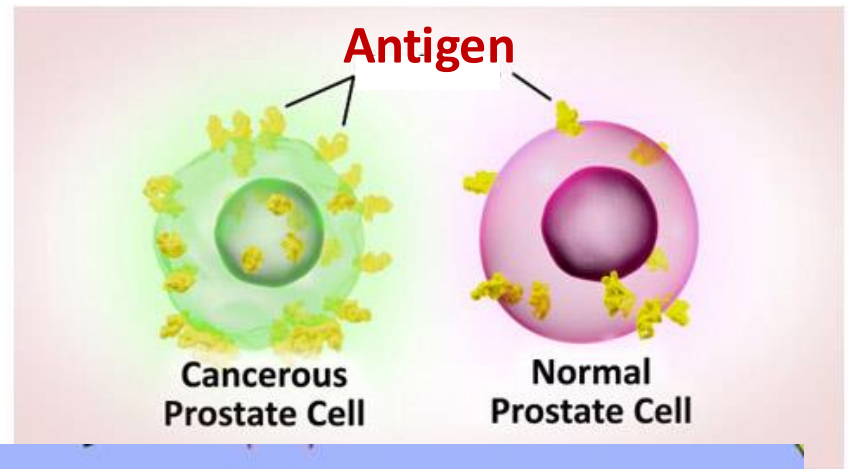
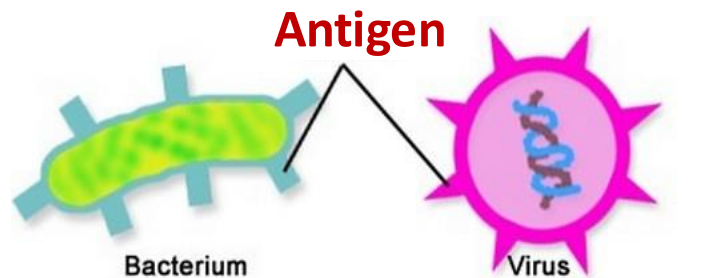
Erythrocytes 5

Antigens

Molecule on a cell that triggers immune cells in a host.

Antigens include:

1. Molecules on surface of bacteria, virus, virally infected cells, cancer cells
2. Toxins secreted by bacteria
3. Molecules on cells of tissue grafts from a donor



Immune System Gone Awry

Autoimmunity: attack “self” cells

- Type I diabetes – attacks pancreatic beta cells
- Rheumatoid arthritis – attacks joints
- Graves disease – attacks TSH receptor → ON → hyperthyroid
- Hashimoto’s disease – attacks thyroid gland → hypothyroid

Allergy: react to nonpathogenic substances

- Food/peanut, insect venom/bee, pollen, red meat, etc.

Immune deficiency: immune system not functioning well

- Weakened immunity: cancer treatments, malnutrition, viral infection (*AIDS: acquired immune deficiency disease/HIV*)
- Genetic defect: congenital; “boy in a bubble”



“Boy in a bubble”

- Severe combined immune deficiency (SCID)
- 6 y.o. NASA created a spacesuit
- Died at 12
- Now treat with bone marrow transplant

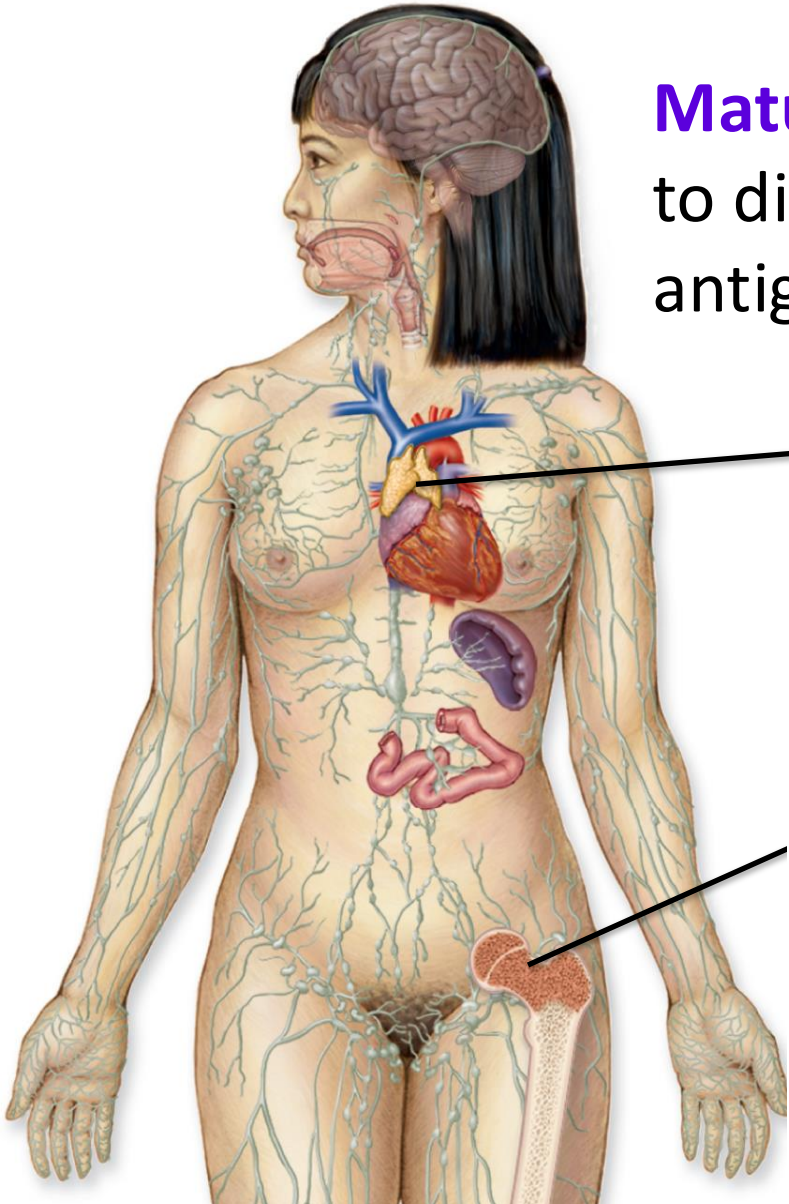
Immune System 1 & 2

1. Introduction
2. Organs & tissues
3. Nonspecific immunity / innate (*born with it*)
4. Overview of cells & chemicals (*mediators*)
5. Specific immunity / adaptive (*acquired*)
6. Immunity & health

Primary Lymphatic Organs

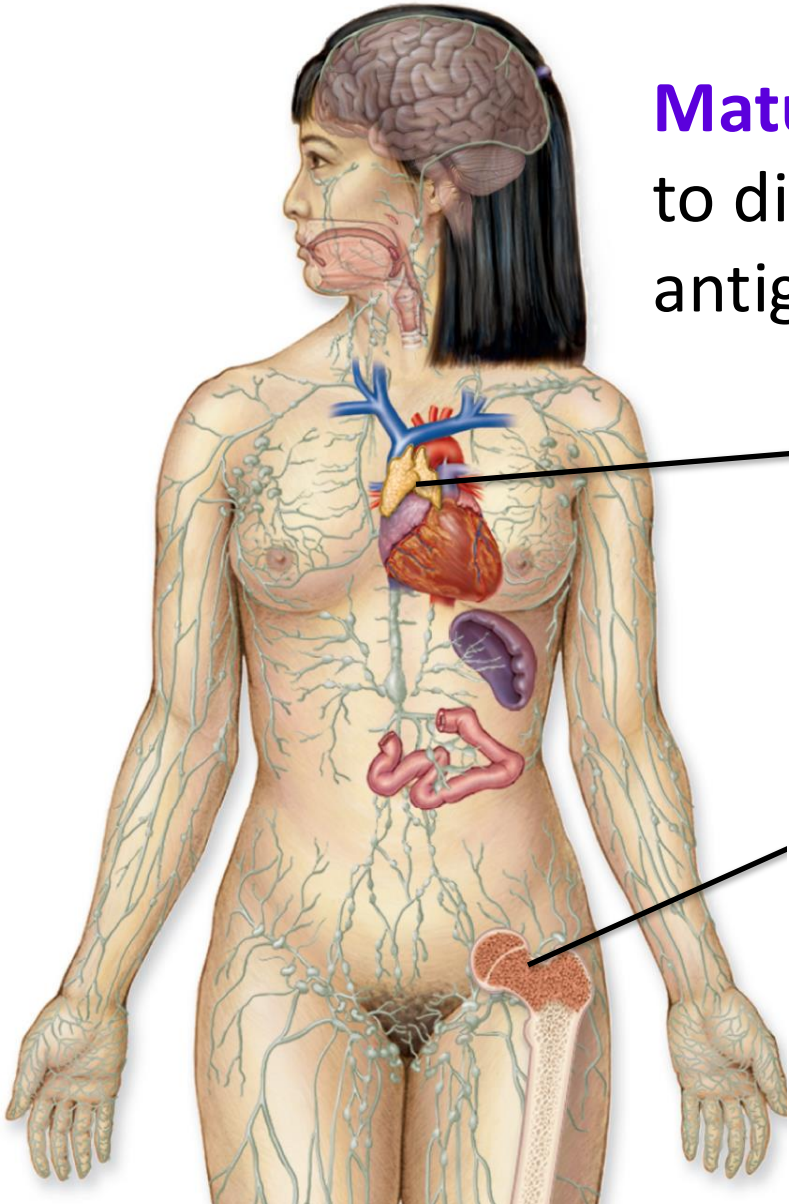
site of production & maturation of WBCs

Maturation: B & T lymphocytes learn to distinguish “self” from “non-self” antigens = immunocompetent.



Thymus

- *T lymphocyte maturation*



Red bone marrow

- *Produces all WBCs (lymphocytes, eosinophils, neutrophils, basophils, monocytes)*
- *B lymphocyte maturation*

Secondary Lymphatic Organs

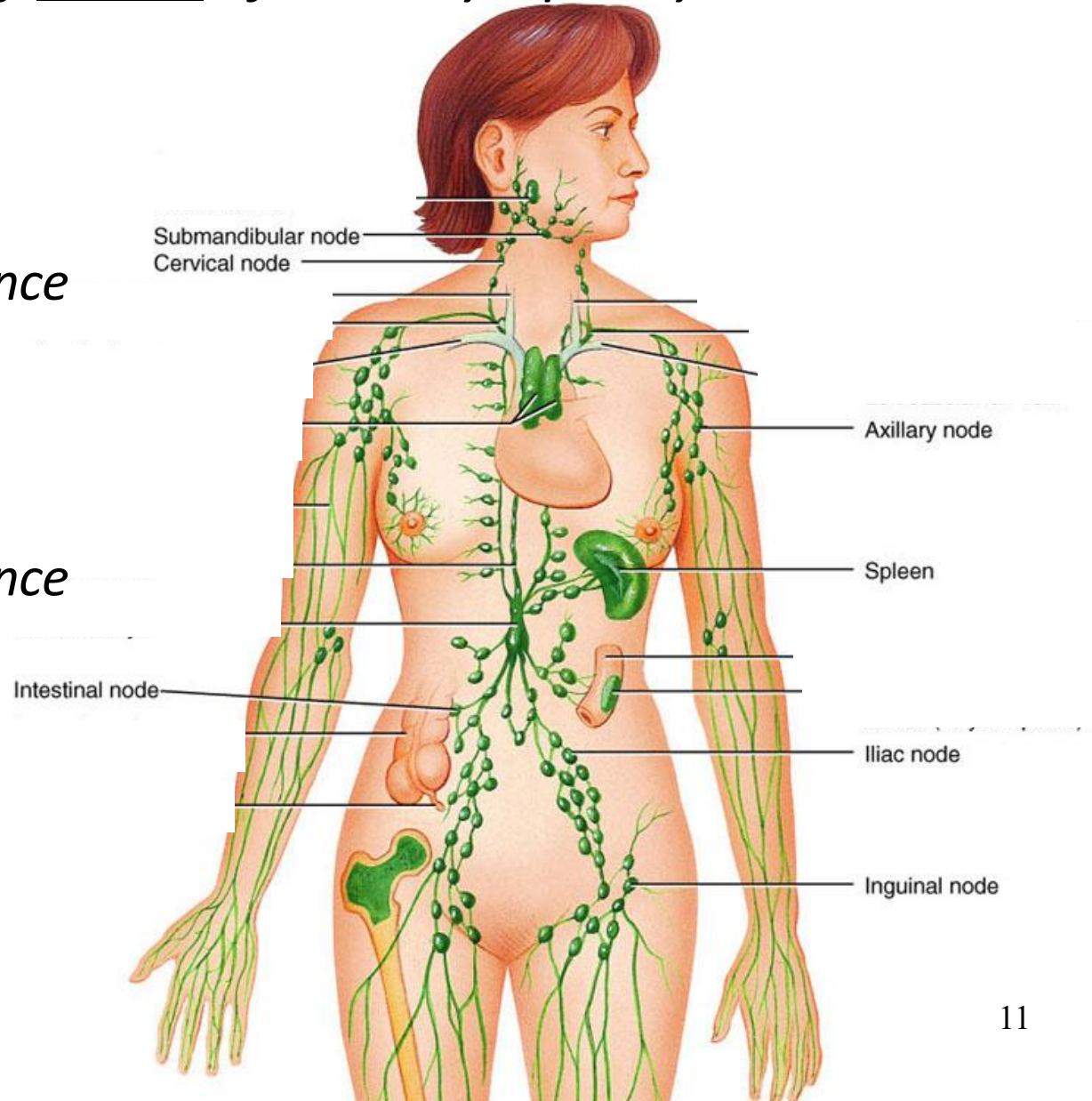
site of action of B & T lymphocytes

Lymph nodes

- *Immune surveillance of lymph*

Spleen

- *Immune surveillance of blood*

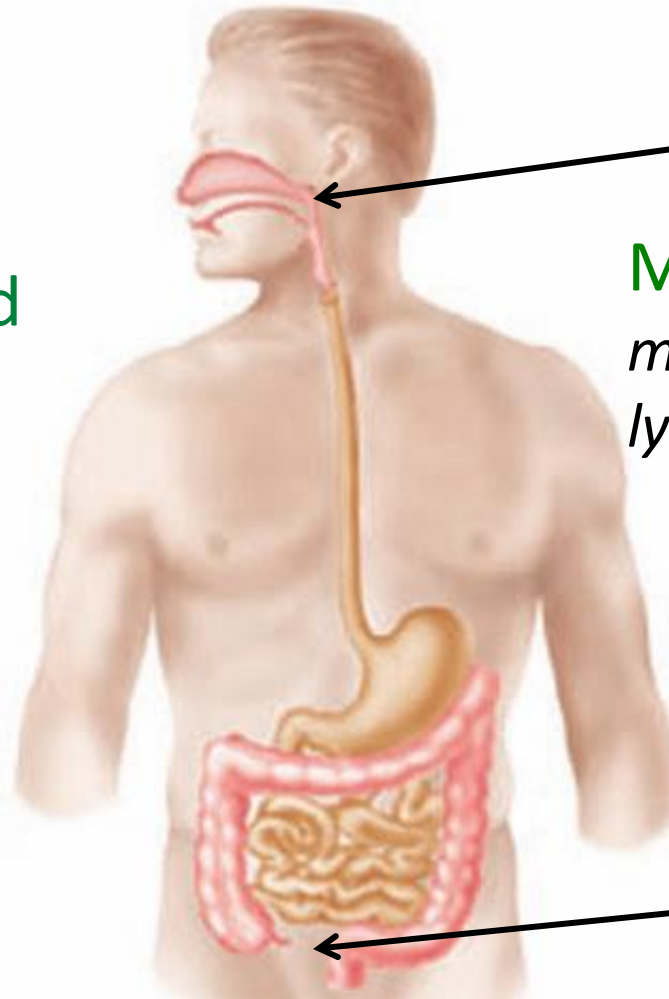


Lymphatic Tissues

site of action of B & T lymphocytes

Found anywhere body
opens to outside world

- *Immune surveillance
mucous membranes*



Tonsils

MALT:

*mucosa associated
lymphoid tissue*

Appendix

NOTE: nonspecific WBCs circulate in blood, until recruited to site of infection

Immune System 1 & 2

1. Introduction
2. Organs & tissues
3. Nonspecific immunity / innate (*born with it*)
4. Overview of cells & chemicals (*mediators*)
5. Specific immunity / adaptive (*acquired*)
6. Immunity & health

Nonspecific Immune System

1. External barriers

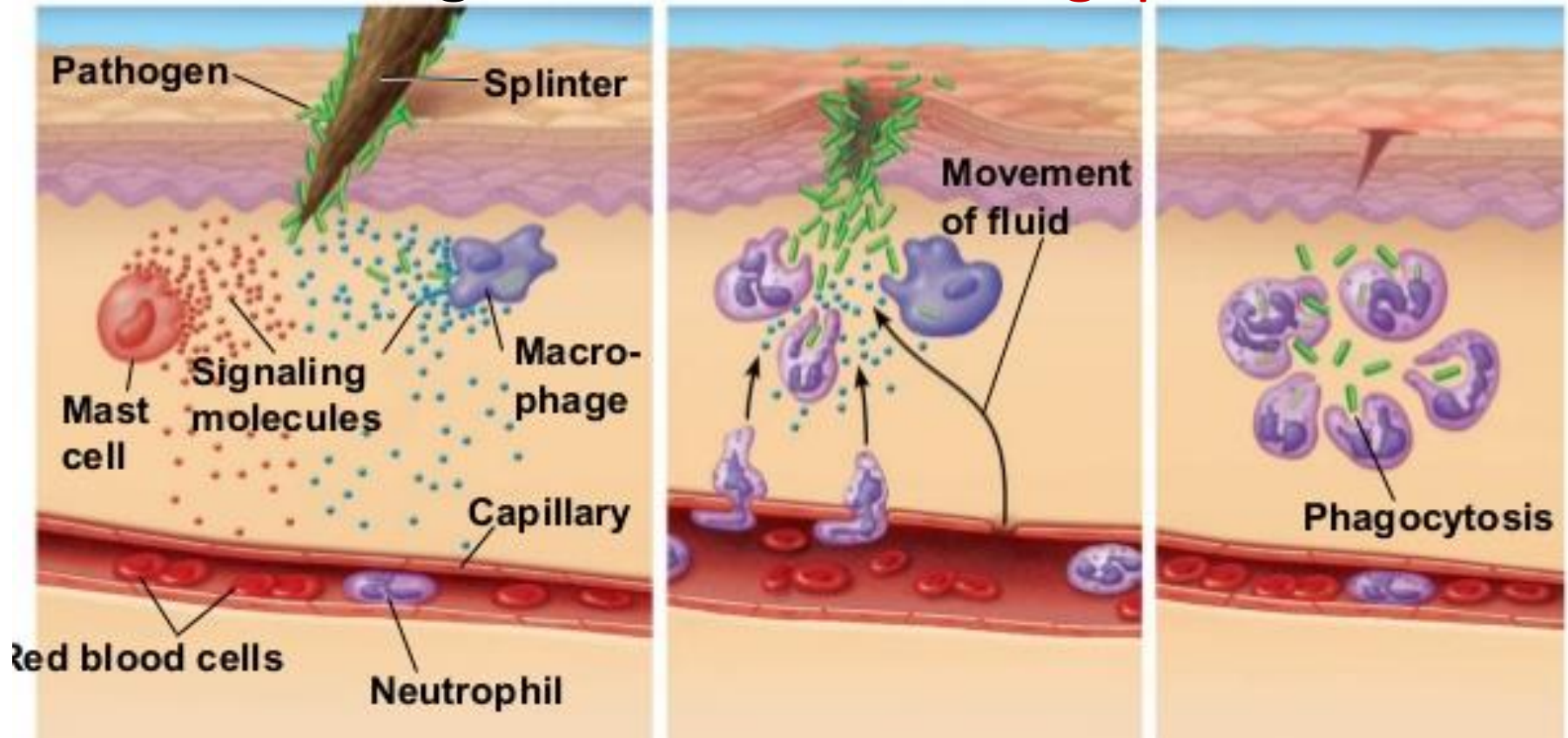
- Cutaneous membrane
- Mucous membranes
- Chemicals in saliva and tears (lysozyme)
- Stomach acid
- Microbiome – healthy microbes
(skin, body cavities open to external environment)

2. Internal barriers

- Chemicals: lysozyme, interferon, complement
- Nonspecific cells: phagocytes, NK cells, eosinophils
- Process: **inflammation**

Nonspecific process: *inflammation*

4 signs: red, hot, swelling, pain



- Splinter damages **mast cells**.
- **Mast cells** release histamine & other chemicals that initiate **inflammation**
- Histamine → **vasodilation**

- Other chemicals: **vessel leakiness**
- **Increase in** neutrophils arriving → leave leaky vessels → phagocytize

Increase blood flow:

- Phagocytes → destroy
- Complement → destroy
- Nutrients → repair
- Heat kills bacteria

Immune System 1 & 2

1. Introduction
2. Organs & tissues
3. Nonspecific immunity / innate (*born with it*)
4. Overview of cells & chemicals (*mediators*)
5. Specific immunity / adaptive (*acquired*)
6. Immunity & health

TABLE: Overview of Cells & Chemicals

of immune system

- **Cells:** categorized by function
- **Chemical mediators:** described by source and how they are used in immunity

Job	White Blood Cell	Notes
signaling cells	mast cells (<i>in tissues</i>)	release histamine
	basophils (<i>in blood</i>)	release histamine
antigen presenting cells	macrophages (<i>in tissue</i>)	present antigen to WBCs
phagocytes	neutrophil (<i>in blood</i>)	1 st to migrate to tissue site of infection
	monocyte (<i>in blood</i>)	become tissue macrophage
	macrophage (<i>in tissue</i>)	monocyte migrate → tissue
lymphocytes	B lymphocyte (<i>B cell</i>) (<i>specific</i>)	mature into plasma cells → produce antibodies
	T lymphocytes (<i>T cells</i>) (<i>specific</i>) T _h : helper T cells (<i>CD4 cells</i>) T _c : cytotoxic (<i>killer</i>) T cells (<i>CD8 cells</i>) T _{reg} : regulatory T cells	enhance B/T cell function destroy affected body cells inhibit B/Tc cells
	NK cells (<i>natural killer</i>)	like cytotoxic T cells/nonspecific
Other	eosinophil	release chemicals toxic to parasitic worms

Chemicals	cell source (<i>made by...</i>)	Notes
histamine	<i>mast cells/ basophils</i>	initiate inflammation; vessels dilate and leak
lysozyme	<i>macrophages</i>	antibacterial enzyme
complement	<i>liver cells; plasma proteins</i>	kills bacteria directly using “membrane attack complex”
interferon	<i>macrophage + virally infected host cells</i>	inhibit viral replication in host cells
cytokines	<i>macrophages, lymphocytes, neutrophils</i> (e.g. interleukin, tumor necrosis factor)	messenger molecules; communicate between WBCs; stimulate proliferation/activation of immune cells
antibodies <i>immunoglobulins</i> (Ig: A, D, E, G, M)	<i>plasma cells</i>	act as B cell receptors + facilitate killing
perforin	<i>killer T cells + NK cells</i>	forms pore in cancer/virally infected cell → death
<i>MHC (major histocompatibility complex)</i>	<i>self cells; plasma membrane proteins</i>	act as identity “tags” on cells; prevent autoimmunity; causes ¹⁹ tissue rejection